



REPUBLIC OF KENYA

NATIONAL OCCUPATIONAL STANDARD

FOR

COMPUTER SCIENCE TECHNICIAN

KNQF LEVEL 6

(CYCLE 3)

PROGRAMME ISCED CODE: 0613 554 A.



**TVET CDACC
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NAIROBI**

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FOREWORD

The provision of quality education and training is fundamental to the Government's overall strategy for social and economic development. Quality education and training contribute to the achievement of Kenya's development blueprint and sustainable development goals.

Reforms in the education sector are necessary to achieve Kenya Vision 2030 and meet the provisions of the Constitution of Kenya 2010. The education sector had to be aligned to the Constitution, and this resulted in the formulation of the Policy Framework for Reforming Education and Training in Kenya (Sessional Paper No. 14 of 2012). A key feature of this policy is the radical change in the design and delivery of TVET training. This policy document requires that training in TVET be competency-based, curriculum development be industry-led, certification be based on demonstration of competence, and the mode of delivery allow for multiple entry and exit in TVET programmes.

These reforms demand that Industry takes a leading role in curriculum development to ensure the curriculum addresses its competence needs. It is against this background that this curriculum has been developed. For trainees to build their skills on foundational hands-on activities of the occupation, units of learning are grouped in modules. This has eliminated duplication of content and streamlined exemptions based on skills acquired as a trainee progresses in the up-skilling process, while at the same time allowing trainees to be employable in the shortest time possible through the acquisition of part qualifications.

It is my conviction that this Occupational Standard will play a great role in developing competent human resources for the Computer Science Sector's growth and development.

PRINCIPAL SECRETARY

STATE DEPARTMENT FOR TVET

MINISTRY OF EDUCATION

PREFACE

Kenya Vision 2030 aims to transform Kenya into a newly industrializing middle-income country, providing high-quality life to all its citizens by the year 2030. Kenya intends to create globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through lifelong education and training. TVET has a responsibility to facilitate the process of inculcating knowledge, skills, and worker behaviour necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency-Based Education and Training (CBET).

CAP 210A and Sessional Paper No. 1 of 2019 on Reforming Education and Training in Kenya for Sustainable Development emphasized the need to reform curriculum development, assessment, and certification. This called for a shift to CBET to address the mismatch between skills acquired through training and skills needed by industry, as well as increase the global competitiveness of the Kenyan labour force.

This Occupational Standards has been developed in adherence to the Kenya National Qualifications Framework and CBETA standards and guidelines. The Occupational Standards is designed and organized into Units of Learning with Learning Outcomes, suggested delivery methods, learning resources, and methods of assessing the trainee's achievement. In addition, the units of learning have been grouped in modules to concretize the skills acquisition process and streamline upskilling.

I am grateful to all expert trainers and everyone who played a role in development of these Occupational Standards.

**CHAIRMAN
TVET CDACC**

ACKNOWLEDGMENT

This Occupational Standards has been designed for competency-based training and has independent units of learning that allow the trainee flexibility in entry and exit. In developing the curriculum, significant involvement and support were received from expert trainers, institutions and organizations.

I recognize with appreciation the role of the ICT National Sector Skills Committee (NSSC) in ensuring that competencies required by the industry are addressed in the curriculum. I also thank all stakeholders in the ICT sector for their valuable input and everyone who participated in developing this Occupational Standards.

I am convinced that this Occupational Standards will go a long way in ensuring that individuals aspiring to work in the ICT Sector acquire competencies to perform their work more efficiently and effectively.

CHAIRPERSON

ICT SECTOR SKILLS ADVISORY COMMITTEE

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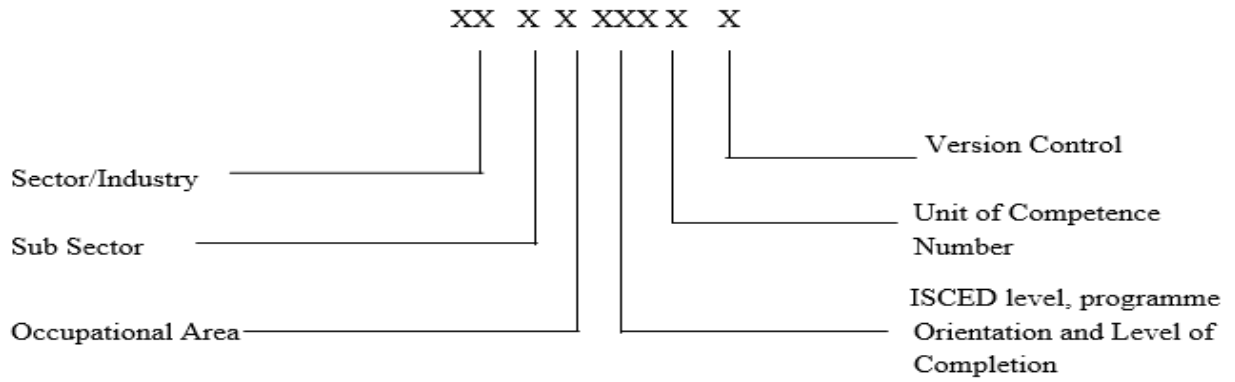
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ABBREVIATIONS AND ACRONYMS

AIDS	Acquired Immunodeficiency Syndrome
BC	Basic Competency
CBET	Competency Based Education and Training
CC	Common Competency
CDACC	Curriculum Development Assessment Certification Council
CEO	Council Secretary
CPU	Central Processing Unit
CR	Core Unit
HIV	Acquired Immunodeficiency Virus
ICT	Information Communication Technology
MA	Control version
OS	Occupational Standard
OSH	Occupational Safety and Health
PESTEL	Political Environmental Social Technological Economic Legal
PPE	Personal Protective Equipment
SOP	Standard Operating Procedure
SSAC	Sector Skills Advisory Committee
SWOT	Strength Weakness Opportunity Threat
TVET	Technical and Vocational Education and Training

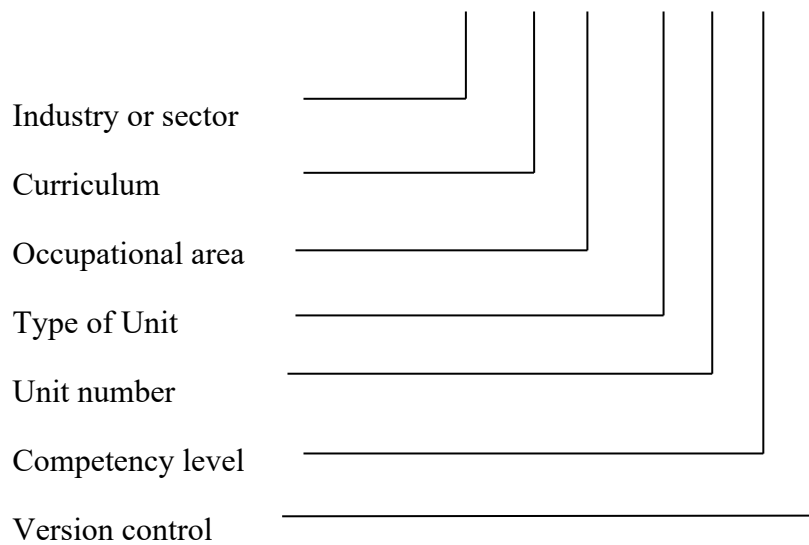
KEY TO UNIT CODE

KEY TO ISCED UNIT CODE



KEY TO TVET CDACC UNIT CODE

ICT /OS /CS /CR / 06 / 6 /MA



OVERVIEW

Computer Science Level 6 qualification consists of competencies that an individual must achieve to offer computer science services. It involves Computer organization and architecture, operating systems, Mathematics for computer science, Fundamentals of programming, Database management skills, Information system, Networking and distributed systems, Artificial intelligence, Algorithms and data structures, Web design skills and Graphic design

Summary of Units of Learning

BASIC UNITS OF COMPETENCY		
ISCED UNIT CODE	TVET CDACC UNIT CODE	UNIT TITLE
0031 441 01A	ICT/OS/CS/BC/02/6/MA	Apply Communication Skills
0417 441 02A	ICT/OS/CS/BC/03/6/MA	Apply Work Ethics and Practices
0413 441 03A	ICT/OS/CS/BC/04/6/MA	Apply Entrepreneurial Skills
COMMON UNITS OF COMPETENCY		
0613 554 07A	ICT/CU/CS/CC/01/6/MA	Apply Basic Electronics
0613 554 08A	ICT/OS/CS/CC/02/6/MA	Apply Fundamentals of Programming
0613 554 10A	ICT/OS/CS/CC/03/6/MA	Apply Mathematics for Computer Science
0613 554 12A	ICT/OS/CS/CR/04/6/MA	Apply Artificial Intelligence Concepts
CORE UNITS OF COMPETENCY		
0613 554 01A	ICT/OS/CS/CR/01/6/MA	Analyze Computer Organization and Architecture
0613 554 02A	ICT/OS/CS/CR/02/6/MA	Configure Operating Systems
0613 554 03A	ICT/OS/CS/CR/03/6/MA	Create Networks and Distributed Systems
0613 554 04A	ICT/OS/CS/CR/04/6/MA	Design Graphics
0613 554 05A	ICT/OS/CS/CR/05/6/MA	Manage Databases
0613 554 06A	ICT/OS/CS/CR/06/6/MA	Design Web Applications
0613 554 09A	ICT/OS/CS/CR/07/6/MA	Create Algorithms and Data Structures

0613 554 11A	ICT/OS/CS/CR/08/6/MA	Develop Information System
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BASIC UNIT OF COMPETENCY

APPLY COMMUNICATION SKILLS

ISCED UNIT CODE: 0031 441 01A

UNIT CODE: CT/OS/CS/BC/02/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to demonstrate communication skills. It involves applying communication channels, written, non-verbal, oral, group communication skills job entry techniques.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT These describe the key outcomes that make up workplace function	PERFORMANCE CRITERIA These are assessable statements that specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in the Range</i>
1. Apply communication channels	1.1 Specific communication channels are identified and applied based on workplace requirements. 1.2 Challenges are identified and addressed as per the operational standards of the organization. 1.3 Communication channels are evaluated to meet workplace needs.
2. Apply written communication skills	2.1 Types of written communication are identified and applied according to the workplace requirements. 2.2 Written communication needs are identified and implemented according to workplace procedures. 2.3 Written communication guidelines are analyzed, evaluated, and revised based on workplace needs.
3. Apply non-verbal communication skills	3.1 Existing non-verbal communication techniques are identified and applied based on organization policy. 3.2 Non-verbal communication techniques are articulated to enhance inclusivity according to workplace requirements. 3.3 Non-verbal communication techniques are modeled to enhance inclusivity according to workplace requirements.
4. Apply oral communication skills	4.1 Types of oral communication are identified and established as per organization policy. 4.2 Pathways of oral communication are identified and established as per organization policy. 4.3 Pathways of oral communication are reviewed according to organization procedures.

ELEMENT These describe the key outcomes that make up workplace function	PERFORMANCE CRITERIA These are assessable statements that specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in the Range</i>
	4.4 Pathways of oral communication are maintained according to the organization standards.
5. Apply group communication skills	5.1 Group communication strategies are applied based on the workplace needs. 5.2 Groups are organized in accordance with workplace procedures. 5.3 Effective questioning, listening and non-verbal communication techniques are used as per needs. 5.4 Group communication challenges are identified and addressed according to the workplace needs.
6. Apply job entry techniques	<i>6.1 Job opportunities</i> are sought based on competencies. <i>6.2</i> A winning resume/CV is developed as per job advertisement. <i>6.3</i> An application/cover letter is developed based on the job advertisement. <i>6.4 certificates and testimonials</i> are organized as per resume. <i>6.5 Interview skills</i> are demonstrated as per job advertisement.

RANGE

This section provides the work environment and conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Communication strategies may include but are not limited to:	● Language switch ● Comprehension check ● Repetition ● Asking confirmation ● Paraphrasing ● Clarification request ● Translation ● Restructuring ● Generalization
2. Effective group interaction may include but not limited to:	● Identifying and evaluating what is occurring within an interaction in a non-judgmental way. ● Using active listening. ● Making decision about appropriate words, behavior. ● Putting together response which is culturally appropriate. ● Expressing an individual perspective.

Variable	Range
	Expressing own philosophy, ideology and background and exploring impact with relevance to communication
3. Situations may include but are not limited to:	- Establishing rapport - Eliciting facts and information - Facilitating resolution of issues - Developing action plans

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required Skills

The individual needs to demonstrate the following skills:

- Active listening
- Interpretation
- Negotiation
- Writing
- Oral skills
- Creative thinking
- Critical thinking
- Decision making
- Analytical
- Innovation
- Conflict skills
- Leadership
- Problem solving skills
- Management
- Organizational
- Teamwork

Required Knowledge

The individual needs to demonstrate knowledge of:

- | | |
|--|--|
| <ul style="list-style-type: none"> Communication process Dynamics of groups Styles of group leadership Key elements of communications strategy Principles of effective communication Turn-taking techniques Conflict resolution techniques Work planning | <ul style="list-style-type: none"> Work organization Company policies Company operations and procedure standards Fundamental rights at the workplace Personal hygiene Accountability Workplace problems and how to deal with them |
|--|--|

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills, knowledge, and range.

1. Critical aspects of Competency.	Assessment requires evidence that the candidate: 1.1 Identified and applied specific communication channels based on workplace requirements. 1.2 Identified and applied specific written communication correspondence according to the workplace requirements. 1.3 Applied and developed non-verbal strategies to communicate in all areas of the workplace requirements. 1.4 Established pathways of oral communication as per workplace policy. 1.5 Applied group communication strategies based on workplace needs. 1.6 Searched for job opportunity based on competencies. 1.7 Prepared job requirement documentations based on job opportunity. 1.8 Demonstrated interview skills based on the job opportunity.
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place. 2.2 Appropriately simulated environment where assessment can take place. 2.3 Resources relevant to the proposed activity or tasks.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Observation 3.2 Oral assessment 3.3 Portfolio of evidence 3.4 Interviews 3.5 Third party report 3.6 Written assessment 3.7 Practical assessment 3.8 Projects
4. Context of Assessment	Competency may be assessed: 4.1 On-the-job 4.2 In a simulated work environment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

APPLY WORK ETHICS AND PRACTICES

ISCED UNIT CODE: 0417 441 02A

UNIT CODE: CT/OS/CS/BC/03/6/MA

UNIT DESCRIPTION

This unit covers competencies required to effectively apply work ethics and practices. It involves the ability to: conduct self-management, promote ethical work practices and values, promote teamwork, manage workplace conflicts, maintain professional and personal development, apply problem-solving and promote customer care.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function.	These are assessable statements which specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in Range</i>
1. Apply self-management skills	1.1 Personal vision, mission and goals are formulated based on potential and concerning organization objectives and strategic plan 1.2 Self-esteem and a positive self-image are developed and maintained based on value 1.3 Emotional intelligence and stress management are demonstrated as per workplace requirements. 1.4 Assertiveness is developed and maintained based on the requirements of the job. 1.5 Accountability and responsibility for one's actions are demonstrated based on workplace instructions. 1.6 Time management, attendance and punctuality are observed as per the organization's policy. 1.7 Personal goals are managed as per the organization's objective 1.8 Self-strengths and weaknesses are identified based on personal objectives 1.9 Motivation, initiative and proactivity are utilized as per the organization policy 1.10 Individual performance is evaluated and monitored according to the agreed targets.
2. Promote ethical work practices and values	2.1 Integrity is demonstrated as per acceptable norms 2.2 Codes of conduct is applied as per the workplace requirements

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function.	These are assessable statements which specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in Range</i>
	2.3 Policies and guidelines are observed as per the workplace requirements 2.4 Professionalism is exercised in line with organizational policies
3. Promote Team work	3.1 Teams are formed to enhance productivity based on organization's objectives 3.2 Duties are assigned to teams under the organization policy. 3.3 Team activities are managed and coordinated as per set objectives. 3.4 Team performance is evaluated based on set targets as per workplace policy. 3.5 Conflicts are resolved between team members in line with organization policy. 3.6 Gender and diversity-related issues are identified and mainstreamed in accordance with workplace policy. 3.7 Healthy relationships are developed and maintained in line with the workplace. 3.8 Adaptability and flexibility are applied in dealing with team members as per workplace policies
4. Maintain professional and personal development	4.1 Personal growth and development needs are identified and assessed in line with the requirements of the job. 4.2 Training and career opportunities are identified and utilized based on job requirements. 4.3 Resources for training are mobilized and allocated based on organizations and individual skills needs. 4.4 Licenses and certifications relevant to the job and career are obtained and renewed as per policy. 4.5 Recognitions are sought as proof of career advancement in line with professional requirements. 4.6 Work priorities and personal commitments are balanced and managed based on the requirements of the job and personal objectives. 4.7 Dynamism and on-the-job learning are embraced in line with the organization's goals and objectives.
5. Apply Problem solving skills	5.1 Creative, innovative and practical solutions are developed based on the problem

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function.	These are assessable statements which specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in Range</i>
	5.2 Independence and initiative in identifying and solving problems are demonstrated based on the requirements of the job. 5.3 Team problems are solved as per the workplace guidelines 5.4 Problem-solving strategies are applied as per the workplace guidelines 5.5 Problems are analyzed and assumptions tested as per the context of data and circumstances
6. Promote Customer Care	6.1 Customers' needs are identified based on their characteristics 6.2 Customer <i>feedback</i> is allowed and facilitated in line with organization policies. 6.3 Customer concerns and complaints are analyzed and resolved in line with the set organizational culture. 6.4 Proactive customer outreach programs are implemented as per organizational policies 6.5 Customer retention strategies are developed and implemented in line with the organizational policy

RANGE

This section provides work environment and conditions to which the performance criteria apply. It allows for different work environment and situations that will affect performance.

Variable	Range
1. Feedback may include but not limited to:	● Verbal ● Written ● Informal ● Formal
2. Conflicts include but are not limited to:	● Interpersonal Conflict. ● Intrapersonal Conflict. ● Intergroup Conflict. ● Intragroup Conflict.

Variable	Range
3. Relationships may include but not limited to:	● Man/Woman ● Trainer/trainee ● Employee/employer ● Client/service provider ● Husband/wife ● Boy/girl ● Parent/child ● Sibling relationships
4. Team may include but not limited to:	● Small work group ● Staff in a section/department ● Inter-agency group ● Virtual teams
5. Personal growth may include but not limited to:	● Growth in the job ● Career mobility ● Gains and exposure the job gives ● Net workings ● Benefits that accrue to the individual as a result of noteworthy performance
6. Personal objectives may include but not limited to:	● Long term ● Short term ● Broad ● Specific
7. Trainings and career opportunities may include but not limited to:	● Participation in training programs ● Serving as Resource Persons in conferences and workshops ● Capacity building
8. Resource may include but not limited to:	● Human ● Financial ● Technology
9. Creative and innovative may include but not limited to:	● New ideas ● Original ideas ● Different ideas ● Methods/procedures ● Processes ● New tools
10. Emerging issues may include but not limited to:	● Artificial Intelligence ● Data confidentiality ● National cohesion ● Open offices

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required Skills

The individual needs to demonstrate the following skills:

- Active listening
- Critical thinking
- Organizational
- Negotiation
- Monitoring
- Evaluation
- Problem solving
- Decision Making
- Leadership
- Creative/innovative thinking
- Adaptability
- Conflict management
- Emotional intelligence
- Teamwork

Required Knowledge

The individual needs to demonstrate knowledge of:

- Work values and ethics
- Company policies and procedures
- Company operations, procedures and standards
- Flexibility and adaptability
- Concept of time and leisure time
- Decision making
- Work planning
- Organizing work
- Monitoring and evaluation
- Record keeping
- Gender and diversity mainstreaming
- Drug and substance abuse
- Professional growth and development
- creativity
- Innovation
- problem solving
- customer care
- mentoring and coaching.
- Emerging issues

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and knowledge and range.

1. Critical aspects of Competency	Assessment require evidence that the candidate: 1.1 Applied self-management skills as per organizational procedures. 1.2 Promoted ethical practices and values as per organizational procedures. 1.3 Promoted Teamwork as per workplace assignments. 1.4 Maintained professional and personal development as per organizational procedures. 1.5 Applied Problem-solving skills based on work requirements. 1.6 Identified customer needs based on their characteristics. 1.7 Gave back Customer feedback in line with organization policies.
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place. 2.3 Resources relevant to the proposed activity or tasks.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Observation 3.2 Oral questioning 3.3 Written test 3.4 Portfolio of Evidence 3.5 Interview 3.6 Third party report
4. Context of Assessment	Competency may be assessed: 4.1 On-the-job 4.2 In a simulated work environment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

APPLY ENTREPRENEURIAL SKILLS

ISCED UNIT CODE: 0413 441 03A

UNIT CODE: CT/OS/CS/BC/04/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to demonstrate an understanding of entrepreneurship. It involves demonstrating an understanding of financial literacy, applying entrepreneurial concepts identifying entrepreneurship opportunities, applying business legal aspects, developing business innovative strategies, and developing business plans.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT These describe the key outcomes that make up workplace function.	PERFORMANCE CRITERIA These are assessable statements that specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in Range</i>
1. Apply Financial Literacy	1.1 Sources of personal and business funds are identified as per financial procedures and standards 1.2 Personal finances are managed as per financial procedures and standards 1.3 Savings are managed as per financial procedures and standards 1.4 Debts are managed as per financial procedures and standards 1.5 Investments are undertaken as per financial procedures and standards 1.6 Insurance services are procured as per financial procedures and standards
2. Apply entrepreneurial concept	2.1 Entrepreneurs and Business persons are distinguished as per principles of entrepreneurship 2.2 <i>Types of entrepreneurs</i> are identified as per principles of entrepreneurship 2.3 Ways of becoming an entrepreneur are identified as per principles of Entrepreneurship 2.4 <i>Characteristics of Entrepreneurs</i> are identified as per principles of Entrepreneurship 2.5 Salaried employment and self-employment are distinguished as per principles of entrepreneurship 2.6 <i>Requirements for entry into self-employment</i> are identified according to business procedures and standards

ELEMENT These describe the key outcomes that make up workplace function.	PERFORMANCE CRITERIA These are assessable statements that specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in Range</i>
	2.7 Roles of an Entrepreneur in an enterprise are determined according to business procedures and standards 2.8 Contributions of entrepreneurship to National development are identified as per business procedures and standards
3. Identify entrepreneurial opportunities	3.1 Business ideas are identified as per business procedures and standards 3.2 Factors to consider when evaluating business opportunity viability are explored based on business procedure and standards 3.3 Entrepreneurial opportunities are evaluated as per business procedures and standards 3.4 Business ideas and opportunities are generated as per business procedures and standards 3.5 Business life cycle is analysed as per business procedures and standards
4. Apply business legal aspects	4.1 <i>Forms of business ownership</i> are identified as per legal procedures and practices 4.2 Business Registration and Licensing processes are identified as per legal procedures and practices 4.3 Types of Contracts and Agreements are analysed as per legal procedures and practices 4.4 Employment Laws are identified as per legal procedures and practices 4.5 Taxation laws are identified as per legal procedures and practices
5. Innovate Business strategies	1. Business innovation strategies are determined by the organization standards 2. Creativity in business development is demonstrated in accordance with business standards 3. <i>Innovative business standards</i> are developed as per business principles 4. Linkages with other entrepreneurs are created as per best practice 5. ICT is incorporated in business growth and development as per best practice
6. Develop Business Plan	6.1 Business idea is described as per business procedures and standards

ELEMENT These describe the key outcomes that make up workplace function.	PERFORMANCE CRITERIA These are assessable statements that specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in Range</i>
	6.2 Business description is developed as per business plan format 6.3 Marketing plan is developed as per business plan format 6.4 Organizational/Management plan is prepared in accordance with business plan format 6.5 Production/operation plan is prepared in accordance with business plan format 6.6 Financial plan is prepared in accordance with the business plan format 6.7 Executive summary is prepared in accordance with business plan format 6.8 Business plan is presented as per best practice 6.9 Business ideas are incubated as per institutional policy.

RANGE

This section provides work environment and conditions to which the performance criteria apply. It allows for different work environment and situations that will affect performance.

Variable	Range
1. Sources of personal funds may include but not limited to:	● Salary/Wages ● Investments ● Savings ● Inheritance ● Government Benefits
2. Sources of business finance may include but not limited to:	● Equity Financing ● Debt Financing, ● Personal Savings/Investment ● Retained Earnings ● Grants and Subsidies ● Crowdfunding ● supplier Credit: ● Leasing and Asset Financing:
3. Types of entrepreneurs may include but not limited to:	● Innovators ● Imitators ● Craft ● Opportunistic ● Speculators

Variable	Range
4. Characteristics of Entrepreneurs may include but not limited to:	• Creative • Innovative • Planner • Risk taker • Networker • Confident • Flexible • Persistent • Patient • Independent • Future oriented • Goal oriented
5. Requirements for entry into self-employment may include but not limited to	• Technical skills • Management skills • Entrepreneurial skills • Resources • Infrastructure
6. Forms of businesses ownership may include but not limited to:	• Sole proprietorship • Partnership • Limited companies • Cooperatives
7. Innovative business standards may include but not limited to:	• New products • New methods of production • New markets • New sources of supplies • Change in industrialization

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required Skills

The individual needs to demonstrate the following skills:

- Analytical
- Management
- Problem-solving
- Root-cause analysis
- Communication

Required Knowledge

The individual needs to demonstrate knowledge of:

- Decision making
- Business communication
- Change management
- Competition
- Risk
- Net working
- Time management
- Leadership
- Factors affecting entrepreneurship development
- Principles of Entrepreneurship
- Features and benefits of common operational practices, e. g., continuous improvement (kaizen), waste elimination,
- Conflict resolution
- Health, safety and environment (HSE) principles and requirements
- Customer care standards
- Basic financial management
- Business strategic planning
- Impact of change on individuals, groups and industries
- Government and regulatory processes
- Local and international market trends
- Product promotion standards
- Market and feasibility studies
- Government and regulatory processes
- Local and international business environment
- Relevant developments in other industries
- Regional/ County business expansion standards

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and knowledge and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified Sources of personal and business finance as per financial procedures and standards 1.2 Managed Personal finances as per financial procedures and standards 1.3 Made Investment decisions as per financial procedures and standards 1.4 Generated Business ideas and opportunities based on business procedure and standards 1.5 Analysed business life cycle based on business procedure and standards 1.6 Determined business innovative standards as per business principles
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	1.7 Developed and presented a business plan as per regulatory framework.
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Written tests 3.2 Oral questions 3.3 Third party report 3.4 Interviews 3.5 Portfolio
4. Context of Assessment	Competency may be assessed: 4.1 On-the-job 4.2 In a simulated work environment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

COMMON UNITS OF COMPETENCY

APPLY BASIC ELECTRONIC SKILLS

ISCED UNIT CODE: 0613 554 07A

UNIT CODE: ICT/OS/CS/CC/01/6/MA

Unit description

This unit specifies the competencies required to apply basic electronics skills. It involves identifying electric circuits and electronic components, understanding semi-conductor theory, identifying and classifying memories, applying number systems and binary coding and identifying emerging trends in electronics.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in the range.</i>
1. Identify electrical circuits	1.1 Electrical components are selected 1.2 <i>Electrical quantities and their units</i> are applied. 1.3 <i>Types of electrical circuits</i> are applied
2. Identify electronic components	2.1 Electrical and electronic components are selected. 2.2 Characteristics of electronic components are applied. 2.3 Application of electronic components are applied 2.4 Characteristics of integrated circuit are applied
3. Apply semi-conductor theory	3.1 Semiconductors are selected 3.2 Structure of matter is applied. 3.3 Electrons in conductors and semiconductors are tested. 3.4 Types of semiconductor materials are applied. 3.5 P-type and N-type materials are designed 3.6 Operations of P-N junction diodes are applied 3.7 Types and <i>operations of transistors</i> are applied.

4. Identify and classify memory	4.1 <i>Types of memories</i> are applied. 4.2 <i>Memory</i> hierarchy is applied 4.3 <i>Classification of memories</i> is applied
5. Apply number systems and binary coding	5.1 <i>Types of number systems</i> are applied 5.2 Types of number systems are applied 5.3 Base conversion is performed. 5.4 Binary arithmetic operations are applied 5.5 Binary codes are applied. 5.6 Representation of decimals in BCD is applied 5.7 BCD arithmetic is performed
6. Identify emerging trends in Electronics	6.1 Emerging trends are applied 6.2 Challenges of emerging trends are applied 6.3 Coping with emerging trends is implemented

RANGE

This section provides work environments and conditions to which the performance criteria apply. It allows for different work environment and situations that will affect performance.

Variable	Range
1. Electrical quantities and their units may include but is not limited to:	E.M.F in volts ● Power in watts ● Energy in joules ● Resistance in ohms ● Current in amperes
2. Types of electrical circuits may include but is not limited to:	● AC – Alternating Current ● DC – Direct Current
3. Types and operations of transistors may include but is not limited to:	● Types ● PNP ● NPN ● Operations ● Forward biasing ● Reverse Biasing
4. Types of memories may include but is not limited to:	● Semi-conductor ● Magnetic ● Optical
5. Levels of memory storage may include but is not limited to:	● Internal ● Main ● Online

Variable	Range
	● Offline bulk
6. Classification of memories may include but is not limited to:	● RAM ● ROM
7. Types of number systems may include but is not limited to:	● Decimal ● Binary ● Octal ● Hexadecimal ● Binary Arithmetic's
8. Binary codes may include but is not limited to:	● 8421 BCD ● Excess 3 ● BCD arithmetic's

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Proficient in ICT
- Time management
- Problem solving
- Decision making
- First aid

Required knowledge

The individual needs to demonstrate knowledge of:

- Electrical Components
- Electrical Quantities and units of measurement
- Electrical circuits
- Semiconductor theory
- Number systems
- Types of Computer memories

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required knowledge and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified Electrical Components, quantities and their units of measurement 1.2 Constructed a simple circuit 1.3 Identified types of transistors and their operations 1.4 Categorized the memories according to their levels, types and hierarchy
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	1.5 Identified the number systems, binary codes and their operations.
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Observation 3.2 Oral questioning 3.3 Practical demonstration
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 On the job 4.3 During industrial attachment
5. Guidance information for assessment	Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

APPLY FUNDAMENTALS OF PROGRAMMING

ISCED UNIT CODE: 0613 554 08A

UNIT CODE: ICT/OS/CS/CR/04/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to understand fundamentals of programming. It involves understanding programming concepts, understanding the Java environment, performing data operations, using control structures, using methods and understanding Object Oriented programming.

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Identify Programming Concepts	1.1 Programming concepts are applied. 1.2 <i>Phases of program development</i> are implemented. 1.3 <i>Key terms used in programming</i> are applied. 1.4 <i>Types of code</i> are executed. 1.5 Program codes are translated 1.6 Fundamentals of OOP are applied
2. Configure the Java environment	2.1 Java is installed 2.2 Java programming environment is configured. 2.3 Features of Java are implemented. 2.4 Java syntax is analyzed.
3. Perform data operations	3.1 <i>Java data types</i> are declared. 3.2 <i>Types of statements</i> are applied 3.3 Variables and constants are declared 3.4 <i>Data operations</i> are implemented. 3.5 Program to perform specified operations is created.
4. Use Control Structures	4.1 <i>Control Structures</i> are implemented. 4.2 Uses of different control statements are employed. 4.3 Programs using control statements are created
5. Use methods	5.1 Procedures/Functions/Methods are applied. 5.2 Methods are implemented. 5.3 Programs using methods are developed
6. Perform Object Oriented Programming	6.1 Object oriented programming is applied 6.2 Classes are applied 6.3 Objects are implemented. 6.4 Inheritance is applied

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
	6.5 Programs involving Inheritance are developed

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Phases of program development may include but not limited to:	● Establish program requirements ● Design a program ● Coding ● Code test and debug ● Document ● Maintain
2. Key terms used in programming may include but not limited to:	● Algorithm ● Source code ● Executable ● Compiling ● Debugging
3. Types of code may include but not limited to:	● Source code ● Object code ● Machine code
4. Java data types may include but not limited to:	● Integer ● Float ● Strings ● Boolean
5. Types of statements may include but not limited to:	● Declaration ● Executable
6. Data Operations may include but not limited to:	● Number operations ● String operations
7. Control Structures may include but not limited to:	● Decision ● Looping

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research

Required knowledge

The individual needs to demonstrate knowledge of:

- Programming concepts
- Compiler operations
- The Java environment
- Data Operations
- Control Structures
- Procedures
- Object Oriented Programming

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1. Explained phases of program development 1.2. Installed Java 1.3. Demonstrated understanding of Java environment 1.4. Created a program to perform data operations 1.5. Explained different types of control statements 1.6. Created a program using control statements 1.7. Created a program using methods 1.8. Explained applications of Object Oriented Programming 1.9. Demonstrated classes and objects 1.10. Demonstrated inheritance
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task

3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical tests 3.3 Observation 3.4 Written test
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

APPLY MATHEMATICS FOR COMPUTER SCIENCE

ISCED UNIT CODE: 0613 554 10A

UNIT CODE: ICT/OS/CS/CR/03/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to understand mathematics for computer science. It involves understanding Linear Algebra, understanding Boolean Algebra, understanding Set Theory, understanding Calculus and understanding Probability and Statistics.

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Apply Linear Algebra	1.1 Linear Equations are applied. 1.2 Linear equations are solved. 1.3 Vectors are applied. 1.4 Vector operations are applied. 1.5 Matrices are applied. 1.6 Matrix operations are applied. 1.7 Inverse of a square matrix is applied.
2. Apply Boolean Algebra	2.1 Boolean algebra is applied. 2.2 Basic Boolean operations are performed. 2.3 Basic boolean operations are applied. 2.4 Secondary Boolean operations are applied 2.5 Writing of Boolean Expressions is implemented. 2.6 Methods of simplifying Boolean expressions are applied. 2.7 Boolean Laws and Theorems are used. 2.8 Simplification rules for Boolean expressions are applied.
3. Apply Set Theory	3.1 Sets Theory is applied. 3.2 Methods of Set representation are used. 3.3 Cardinality of a set determined. 3.4 Types of sets are classified. 3.5 Venn Diagrams are constructed. 3.6 Set Operations are Performed.
4. Apply Calculus	4.1 Functions and graphs are applied. 4.2 Differential calculus is performed. 4.3 Integral calculus is computed.
5. Apply Probability and Statistics	5.1 Business innovation strategies are applied according to organizational standards 5.2 Key terminologies in Probability are applied.

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
	5.3 Probability axioms and simple counting problems are solved. 5.4 Permutations and combinations are computed. 5.5 Conditional probability and the multiplication rule are applied. 5.6 Key terminologies in Probability are applied. 5.7 Data representation techniques are used. 5.8 Measures of central tendency are calculated. 5.9 Measures of spread are calculated. 5.10 Measures of Location are computed.

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Vector operations may include but not limited to:	• Addition • Multiplication • Dot product
2. Matrix operations may include but not limited to:	• Sum of two matrices • Sum of a matrix and a scalar • Matrix subtraction • Product of two matrices • Product of a matrix and a vector
3. Basic Boolean operations may include but not limited to:	• AND • OR • NOT
4. Secondary operations may include but not limited to:	• NAND • NOR • EX-OR • EX-NOR
5. Methods of simplifying Boolean expressions may include but not limited to:	• Using algebraic functions • Using Truth tables • Using Karnaugh Maps
6. Boolean Laws and Theorems may include but not limited to:	• AND law • OR law • Inversion law

Variable	Range
	• Commutative • Associative • Distributive • De-Morgan's Theorems
7. Methods of Set representation may include but not limited to:	• Statement form • Tabular form • Set builder notation
8. Types of sets may include but not limited to:	• Finite Set • Infinite Set • Subset • Proper Subset • Universal Set • Empty or Null • Equal • Equivalent Set • Singleton Set or Unit Set • Overlapping Set • Disjoint Set
9. Set Operations may include but not limited to:	• Set Union and Set Intersection • Set Difference/Relative Complement • Set Complement • Cartesian Product
10. Measures of central tendency may include but not limited to:	• Mean • Median • Mode
11. Measures of spread may include but not limited to:	• Variance • Standard deviation
12. Measures of location may include but not limited to:	• Percentile • Quartiles

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

- Linear Algebra
- Boolean algebra
- Set Theory
- Calculus Probability and Statistics

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Solved Linear equations 1.2 Performed vector operations 1.3 Performed matrix operations 1.4 Performed Boolean algebra operations 1.5 Performed set operations 1.6 Explained samples spaces, events and sets 1.7 Solved problems using Probability axioms 1.8 Solved permutations and combinations 1.9 Solved problems using conditional probability 1.10 Represented data using statistical technique 1.11 Illustrated measures of central tendency 1.12 Illustrated measures of spread 1.13 Illustrated measures of location
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical tests 3.3 Observation 3.4 Written test
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment

5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.
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APPLY ARTIFICIAL INTELLIGENCE CONCEPTS

ISCED UNIT CODE: 0613 554 12A

UNIT CODE: ICT/OS/CS/CC/04/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to understand artificial intelligence. It involves understanding fundamentals of Artificial Intelligence, understanding problem solving techniques, understanding Python programming environment and developing Artificial Intelligence programs using Python.

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Identify fundamentals of Artificial Intelligence	1.1 Fundamentals of artificial intelligence are applied. 1.2 The history of Artificial Intelligence is applied. 1.3 Foundations of Artificial Intelligence are applied. 1.4 Artificial Intelligence applications are implemented. 1.5 Intelligence agents are applied. 1.6 Artificial Intelligence applications in real life are applied.
2. Use problem solving techniques	2.1 Logical operators are applied. 2.2 Propositional and Predicate logic are implemented. 2.3 <i>Types of inferencing</i> are applied. 2.4 Machine Learning is implemented. 2.5 <i>Types of Machine Learning</i> are applied. 2.6 Applications of different types of inferencing are deployed
3. Configure Python programming environment	3.1 Installation of Python is performed. 3.2 Python syntax is applied. 3.3 Python <i>Data types</i> are identified. 3.4 Control structures in Python are applied. 3.5 Functions in python are developed. 3.6 Object Oriented Python is applied. 3.7 <i>Scientific Modules</i> in Python are applied. 3.8 Applications are created using scientific modules
4. Develop Artificial Intelligence programs using python	4.1 Sci-Kit Learn is applied. 4.2 Machine Learning with K-Nearest Neighbours is implemented. 4.3 Machine Learning with Naïve Bayes Algorithm is applied.

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Types of inferencing may include but not limited to:	• Single • Multiple • Case based
2. Types of Machine Learning may include but not limited to:	• Supervised • Unsupervised
3. Data types may include but not limited to:	• Integers • Floats • Strings • Lists • Tuple • Sets • Dictionaries
4. Scientific Modules may include but not limited to:	• Numpy • Pandas • Matplotlib

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

- Concepts of Artificial Intelligence
- Problem solving techniques
- Python programming environment
- Development of Artificial Intelligence programs using python

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Explained applications of artificial intelligence 1.2 Explained the role of intelligence agents 1.3 Explained types of inferencing 1.4 Explained types of machine learning 1.5 Demonstrated installation of Python 1.6 Demonstrated Python syntax 1.7 Demonstrate data types in Python 1.8 Demonstrated use of control structures in Python 1.9 Demonstrated use of functions in Python 1.10 Demonstrated use of Object Oriented Python 1.11 Demonstrated use of scientific modules 1.12 Demonstrated machine learning
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical tests 3.3 Observation 3.4 Written tests
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

CORE UNITS OF COMPETENCY

ANALYSE COMPUTER ORGANISATION AND ARCHITECTURE

ISCED UNIT CODE: 0613 554 01A

UNIT CODE: ICT/OS/CS/CR/01/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to understand Computer Organisation and Architecture. It involves understanding principles of computer organisation and design, understanding central processing unit functions, understanding computer memory functions, understanding input-output functions and understanding computer arithmetic and logic.

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Apply principles of computer organization and design	1.1 ICT concepts are applied in relation to computer equipment 1.2 Principles of computer organisation are applied 1.3 Computer architecture is analysed and documented in terms of system performance 1.4 Structure and function of computer hardware components are identified
2. Analyse central processing unit functions	2.1 Peripheral devices are selected 2.2 The Central Processing Unit (CPU) is selected as per user requirements 2.3 CPU architecture is analyzed 2.4 Registers are analyzed 2.5 Instruction representation and execution are analyzed 2.6 CPU specifications are identified and selected as per user's requirements 2.7 CPU specifications are verified on a given computer system
3. Analyse computer memory functions	3.1 Memory organization is applied 3.2 storage technologies are applied 3.3 Cache and virtual memory are analysed 3.4 Memory specifications are selected as per user's requirements 3.5 Memory specifications are verified on a given computer system
4. Utilize input-output devices	4.1 Peripheral devices are installed 4.2 Input-output processing is analysed

ELEMENT These describe the key outcomes which make up workplace function.	PERFORMANCE CRITERIA These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
	4.3 Bus interface is applied 4.4 <i>Modes of data transfer</i> are analysed 4.5 <i>Input-output device specifications</i> are selected per user's requirements 4.6 Input-output device specifications are verified on a given computer system 4.7 Input-output processing is performed
5. Analyse computer arithmetic and logic	5.1 <i>Number systems</i> are identified and applied 5.2 Integer and floating-point representations are applied based on IEEE standards 5.3 Integer and floating-point arithmetic is applied 5.4 <i>Logic operators</i> are applied 5.5 Logic operations are applied 5.6 <i>Methods of representing logic operations</i> are applied

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Computer hardware components	● Input devices ● Output devices ● Storage devices ● Central Processing unit (CPU)
2. CPU specifications may include but is not limited to:	● Brand ● Chipset ● Speed Series
3. Storage Technologies may include but is not limited to:	● Solid state ● Magnetic ● Optical
4. Memory specifications may include but is not limited to:	● Speed ● Size ● Form factor ● Type Part Number
5. Modes of data transfer may include but is not limited to:	● Programmed I/O ● Direct Memory Access I/O

	● Interrupt initiated I/O
6. Input-output device specifications may include but is not limited to:	● Monitor: Size, Resolution, Brand ● Printer/Copier: Function, Speed, Resolution, Brand ● Storage: Size, Brand, Data Transfer Rate
7. Number systems 8. may include but is not limited to:	● Decimal ● Positional ● Binary ● Hexadecimal
9. Logic Operators may include but is not limited to:	● AND ● OR ● NOT
10. Methods of representing logic operations may include but is not limited to:	● Karnaugh maps ● Logic gates ● Truth tables

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

- Fundamentals of ICT
- Principles of computer organisation and design
- Central Processing Unit functions
- Computer memory functions
- Input-Output functions
- Computer arithmetic and logic

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified Computer organization and architecture 1.2 Analyzed and documented Computer architecture in terms of system performance 1.3 Identified Structure and function of computer components 1.4 Selected Central Processing Unit (CPU) as per user requirements 1.5 Analyzed CPU architecture and register 1.6 Selected and verified memory specifications as per user's requirements 1.7 Analyzed Cache and virtual memory 1.8 Installed Peripheral devices 1.9 Analyzed Input-output processing 1.10 Analyzed modes of data transfer 1.11 1.12 Selected and verified Input-output device specifications as per user's requirements 1.13 Identified and applied Number systems 1.14 Applied integer and floating-point representations based on IEEE standards 1.15 Applied Integer and floating-point arithmetic 1.16 Applied Logic operations
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical tests 3.3 Observation 3.4 Written test
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 On the job 4.3 During industrial attachment
5 Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

CONFIGURE OPERATING SYSTEMS

ISCED UNIT CODE: 0613 554 02A

UNIT CODE: ICT/OS/CS/CR/02/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to understand operating systems. It involves understanding fundamentals of operating systems, applying computer application software to solving tasks, understanding process management, understanding memory management, understanding input-output management and understanding file management.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT	PERFORMANCE CRITERIA <i>(Bold and italicised terms are elaborated in the Range)</i>
1. Identify fundamentals of operating systems	1.1 Principles of computer software are applied 1.2 <i>Types of operating systems</i> are identified and analysed 1.3 Operating system is selected according to manufacturer's specifications 1.4 Operating system types are applied 1.5 <i>Requirements of operating systems</i> are applied 1.6 Installation of Windows is performed
2. Apply computer application software to solving tasks	2.1 <i>Word documents are</i> formatted as per job requirements 2.2 <i>Presentation slides</i> are prepared and presented in accordance with workplace procedures. 2.3 <i>Worksheets and workbooks</i> are prepared as per job requirements 2.4 <i>Database is created, populated, and manipulated</i> in accordance with workplace procedures 2.5 Data manipulation is performed 2.6 Office internet functions are implemented in accordance with office procedures 2.7 Electronic mail addresses are created in accordance to workplace communication office policy
3. Apply process management	3.1 Computer resources are monitored 3.2 Process states and transitions are monitored 3.3 Process scheduling is applied 3.4 Task Manager is utilized as per work requirements 3.5 performance monitor tool is utilized as per work task
4. Manage memory	4.1 Memory management settings are configured. 4.2 <i>Memory management techniques</i> are applied 4.3 Virtual memory management settings are configured

ELEMENT	PERFORMANCE CRITERIA <i>(Bold and italicised terms are elaborated in the Range)</i>
5. Manage input and output devices	5.1 Disk operations are performed 5.2 Computer clock system is monitored 5.3 Disk is selected according to user needs 5.4 Disk management is performed as per user needs 5.5 Removable disks are selected
6. Manage files and configure local policy settings	6.1 <i>File access methods</i> are applied 6.2 File allocation techniques are verified 6.3 File protection and security measures are configured 6.4 <i>File and directory operations</i> are applied 6.5 <i>Local policy settings</i> are configured

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Computer softwares may include but not limited to:	● System software ● Application software ● Utility software
2. Structures of operating system may include but is not limited to:	● Monolithic ● Layered ● Virtual ● Client server model
3. Types of operating system may include but is not limited to:	● Real time ● Normal ● Batch ● Time sharing
4. Word documents preparation may include but not limited to:	● Creation ● Editing ● Formatting ● Mail merging ● Printing
5. Slides preparation may include but not limited to:	● Creation ● Editing ● Formatting ● Printing
6. Worksheets and workbooks preparation may include but not limited to:	● Creation ● Data entry ● Basic formulae and functions

	● Formatting ● Data sorting and filtering ● Printing
7. Database design creation and manipulation may include but not limited to:	● Table design ● Form design ● Report design ● Data sorting ● Indexing ● Storage ● Retrieval/querying ● Security ● Printing
8. Computer Resources may include but is not limited to:	● Processor ● Storage space
9. Computer Resources may include but is not limited to:	● Processor ● Storage space
10. Memory management techniques may include but is not limited to:	● Partitions ● Virtual
11. Disk storage management operations may include but is not limited to:	● Shrinking volume ● Extending volume ● Formatting volume ● Partitioning volume ● Disk Optimization and defragmentation
12. Device Management Operations may include but is not limited to:	● Driver Installation ● Resolving driver conflicts
13. File access methods may include but is not limited to:	● Sequential ● Random ● Indexed sequential
14. File and directory operations may include but is not limited to:	● Setting attributes ● Share settings ● Security settings ● Customization of files and folders
15. Local policy settings may include but is not limited to:	● Password policy ● Account lockout policy ● Audit policy ● Security options

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

Communications (verbal and written);

- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

Classification of computer software

- Word processing;
- Spread sheets;
- Database;
- Presentation Packages;
- Office internet
- Concepts of operating systems
- Process management
- Memory management
- Input/output management
- File management and local security policy settings

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1 Selected and installed computer software according to manufacturer's specifications and user specifications 1.1 Formatted Word documents as per job requirements 1.2 Prepared and presented slides in accordance with workplace procedures. 1.3 Prepared worksheets and workbooks as per job requirements 1.4 Created, populated, and manipulated Database in accordance with workplace procedures 1.5 Created electronic mail addresses in accordance to workplace communication office policy 1.6 Monitored process states and transitions 1.7 Configured memory management settings 1.8 Configured virtual memory management settings 1.9 monitored Computer clock system 1.10 Performed virtual input/output operations in relation to system devices 1.11 Verified Disk properties 1.12 Performed disk storage management operations
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	1.13 Verified file allocation techniques 1.14 Configured file protection and security measures 1.15 Applied file and directory operations 1.16 Configured local policy settings
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral test 3.2 Observation 3.3 Practical demonstration 3.4 Written tests
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

CREATE NETWORKS AND DISTRIBUTED SYSTEMS

ISCED UNIT CODE: 0613 554 03A

UNIT CODE: ICT/OS/CS/CR/07/6/MA

UNIT DESCRIPTION:

This unit specifies the competencies required to understand networking and distributed systems concepts. It involves understanding networking and distributed systems, distributed system architectures, distributed processing and file management, setting up a network in a distributed environment, understanding data communication standards and IP addressing and troubleshooting a network.

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Apply networking and distributed systems concepts	1.1 Fundamentals of networking are applied 1.2 <i>Types of networks</i> are selected 1.3 <i>Network topologies</i> are applied 1.4 Transmission media are terminated 1.5 Distributed system is configured 1.6 <i>Types of distributed systems</i> are applied 1.7 <i>Models in distributed systems</i> are chosen 1.8 Network requirements are specified 1.9 Network security is configure based on local threats 1.10 Components of network security are applied 1.11 Wireless security is applied
2. Create distributed systems architectures	2.1 Distributed architecture is selected and created 2.2 <i>Architecture styles</i> are applied 2.3 Classification of <i>types of distributed system architectures</i> are applied 2.4 Distributed system architecture requirements for a simulated site are analyzed
3. Create distributed processing and file management	3.1 <i>Types of distributed processing</i> are applied 3.2 File systems are implemented 3.3 <i>File sharing and accessing methods</i> are implemented 3.4 Distributed file sharing and access are implemented
4. Set up network in distributed environment	4.1 Tools, materials and devices for network set up are assembled according to the network type

ELEMENT These describe the key outcomes which make up workplace function .	PERFORMANCE CRITERIA These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
	4.2 Network devices are connected and configured according to local and international standards 4.3 Network software is installed and configured according to the user manual 4.4 Network performance is tested
5. Apply Data Communication Standards and IP addressing	5.1 OSI Model is applied 5.2 Data communication components are applied 5.3 Network IP address classes are configured
6. Troubleshoot a network	6.1 Troubleshooting techniques are applied 6.2 Troubleshooting tools are selected. 6.3 Troubleshooting is performed.

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Types of networks may include but not limited to:	• LAN • WAN • MAN • PAN
2. Network topologies may include but not limited to:	• Bus • Star • Delta • Ring • mesh point-to-point
3. Types of distributed systems may include but not limited to:	• Computing • Information • Pervasive
4. Models in distributed 5. systems may include but not limited to:	• Architecture • Interaction • Fault
6. Architecture styles may include but not limited to:	• Layered Architecture • Object Based Architecture • Data-centered Architecture • Event Based Architecture • Hybrid Architecture

Variable	Range
7. Types of distributed system architecture may include but not limited to:	● Centralized ● Decentralized ● Hybrid
8. Types of distributed processing	● Distributed ● Parallel
9. File sharing and access methods may include but not limited to:	● Remote Access ● Data-Caching
10. Troubleshooting tools may include but not limited to:	● Ping ● Tracert / traceroute ● Nslookup ● Netstat ● Pathping/mtr

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research

Required knowledge

The individual needs to demonstrate knowledge of:

- Fundamentals of networking and distributed systems
- Distributed systems architectures
- Distributed processing and file management
- Setting up a network in a distributed environment
- Troubleshooting a network

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Analyzed network requirements for a site 1.2 Selected network as per user requirement 1.3 Applied Network topologies 1.4 Terminated transmission media
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	1.5 Configured distributed system 1.6 Selected and created distributed architecture 1.7 Analyzed distributed system architecture requirements for a simulated site 1.8 Created file sharing and accessing methods 1.9 Carried out distributed file sharing and access 1.10 Connected and configured Network devices according to local and international standards 1.11 Installed and configured Network software according to the user manual 1.12 Tested Network performance 1.13 Outlined OSI Model 1.14 Configured Data communication components 1.15 Configured Network IP address classes 1.16 Performed Troubleshooting of network as per IEEE standards
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral tests 3.2 Observation 3.3 Practical demonstration 3.4 Written tests
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

DESIGN GRAPHICS

ISCED UNIT CODE: 0613 554 04A

UNIT CODE: ICT/OS/CS/CR/11/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to understand Graphic Design. It involves applying fundamentals of graphic design, applying elements and principles of graphic design, applying typography techniques, creating and editing of images, performing layout design and printing the design.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT	PERFORMANCE CRITERIA <i>(Bold and italicised terms are elaborated in the Range)</i>
1. Apply fundamentals of graphic design	1.1 Graphics design fundamentals are applied 1.2 Graphic design equipment is selected based on the design. 1.3 Graphic design user requirements are analyzed 1.4 Applications areas of Graphic design is explored.
2. Apply elements and principles of graphic design	2.1 Elements of graphic design project are selected as per user requirements 2.2 Principles of graphic design project are applied as per user requirements 2.3 Elements of graphic design are applied
3. Apply typography techniques	3.1 Typography guidelines are applied 3.2 Applied measurements and standards of typography. 3.3 Typography technique for a graphic design is selected and applied project as per user requirements.
4. Create and edit images	4.1 software and tools for graphic design and photography are identified. 4.2 Image file types are identified 4.3 Letter forms, lines of type and body copy are created 4.4 Images are created and manipulated
5. Perform layout design	5.1 Applied proportion on layout design. 5.2 Unified systems are created of out dissimilar elements. 5.3 Dynamic layouts are created by using typographic tools 5.4 Created type and image project. 5.5 Multi-page layout planning was performed

ELEMENT	PERFORMANCE CRITERIA <i>(Bold and italicised terms are elaborated in the Range)</i>
6. Print design	6.1 Tools and equipment for printing are selected 6.2 <i>Types of printing</i> are selected and applied based on the design. 6.3 Paper is classified according to types, size and weight. 6.4 Chemicals used in printing are selected 6.5 Printed actual design

RANGE

This section provides work environment and conditions to which the performance Criteria apply. It allows for different work environment and situations that will affect performance.

Variable	Range
1. Graphic design equipment may include but not limited to:	● Computer ● Scanner ● Printer ● Camera ● Digital Tablet
2. Image file types may include but not limited to:	● Raster ● Vector
3. Typographical tools may include but not limited to:	● Microsoft Publisher ● Illustrator ● Adobe InDesign ● Adobe Photoshop ● Paint.net ● Corel Draw
4. Types of printing may include but not limited to:	● Digital ● Flexography ● Letterpress ● Off set ● Rotogravure ● Screen

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

- Fundamentals of graphic design
- Elements and principles of graphic design
- Typography techniques
- Creating and editing Images
- Layout Design
- Printing graphics

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Determined graphic design user specifications 1.2 Selected and applied elements of graphic design project as per user requirements 1.3 Applied Measurements, standards and guidelines of typography. 1.4 Selected and applied typography technique for a graphic design project as per user requirements. 1.5 Identified image file types 1.6 Created letter forms, lines of type and body copy 1.7 Created and manipulated images based on software 1.8 Created unified systems of out dissimilar elements. 1.9 Selected and applied types of printing based on the design
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task

3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical tests 3.3 Observation 3.4 Written tests
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

MANAGE DATABASES

ISCED UNIT CODE: 0613 554 05A

UNIT CODE: ICT/OS/CS/CR/05/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to demonstrate database management skills. It involves understanding database fundamentals, designing a database, using Structured Query Language, understanding design of object oriented databases, understanding indexing and hashing and understanding database applications.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT These describe the key outcomes which make up workplace function .	PERFORMANCE CRITERIA These are assessable statements which specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in the range.)</i>
1. Apply Database fundamentals	1.1 Database is created as per design. 1.2 <i>Terminologies used with databases</i> are applied 1.3 Benefits of using databases are applied 1.4 Relational database Model is applied 1.5 Key concepts in relational modelling are applied. 1.6 Properties of a table/relation are utilised 1.7 Relational Database Management Systems (RDBMSs) products is selected 1.8 MS SQL server is installed 1.9 MS SQL server interface is Navigated and utilized 1.10 <i>Properties of MS SQL server</i> database are applied 1.11 RDBMS is applied 1.12 Database security is implemented
2. Design a database	2.1 <i>Phases of database design</i> is applied 2.2 Entity modeling using UML notation is constructed 2.3 Database designs is normalized 2.4 ER model is validated according to the requirements
3. Use Structured Query Language	3.1 Structured Query Language (SQL) is applied 3.2 <i>Data definition queries</i> are applied 3.3 Created tables using the SQL CREATE TABLE statemen are created 3.4 <i>CREATE TABLE statement constraints</i> is applied 3.5 Table schema using the SQL ALTER statement is edited

ELEMENT These describe the key outcomes which make up workplace function .	PERFORMANCE CRITERIA These are assessable statements which specify the required level of performance for each of the elements. <i>Bold and italicized terms are elaborated in the range.)</i>
	3.6 Table is dropped using the SQL DROP TABLE statement 3.7 Data <i>manipulation query statement</i> is applied 3.8 Data is manipulated using SQL statements 3.9 SQL join is utilised 3.10 Database from validated ER model is created and queried 3.11 <i>SQL joins</i> are utilised
4. Design an object oriented databases	4.1 Object oriented database is applied 4.2 Applied <i>Object oriented database concepts</i> is applied 4.3 Object Oriented database concepts from a set of requirements is applied 4.4 <i>Triggers</i> and views in object oriented databases are created.
5. Index and hash	5.1 Indexing and hashing is applied 5.2 Index in databases is created 5.3 Hashing in databases is applied 5.4 Indexing and hashing in an existing database is implemented
6. Design Database applications	6.1 Decision support systems is applied 6.2 Applied Data mining is applied 6.3 Distributed databases is created 6.4 Data is warehoused 6.5 Spatial and geographical databases features are applied 6.6 Multi-media databases are utilised 6.7 Mobility and personal databases are applied 6.8 Data warehouses from a given set of requirements is designed

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Terminologies used with databases may include but not limited to:	● Table ● Records ● Field

Variable	Range
	● DBMS
2. Properties of MS SQL server database may include but not limited to:	● Deleting a database ● Deleting data or log files ● Increasing database size ● Shrinking database ● Renaming database ● Importing a database ● Exporting a database
3. Phases of database design may include but not limited to:	● Conceptual design ● Logical design ● Physical design
4. Data definition queries may include but not limited to:	● CREATE ● DROP ● ALTER
5. CREATE TABLE statement constraints may include but not limited to:	● Primary key ● Foreign key ● UNIQUE ● CHECK ● NOT NULL ● DEFAULT
6. Data manipulation query statements may include but not limited to:	● INSERT ● SELECT ● UPDATE ● DELETE
7. Types of joins may include but not limited to:	● Simple Join or Inner Join ● Left Join ● Right Join ● Outer Join
8. Object oriented database concepts may include but not limited to:	● Classes ● Objects ● Attributes ● Inheritance
9. Views may include but not limited to:	● Create a view ● Rename a view ● Drop a view
10. Triggers may include but not limited to:	● Create a trigger ● Alter a trigger ● Drop a trigger

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

- Database concepts
- Database design
- Structured Query Language
- Object oriented database design
- Applications of object oriented databases

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Installed MS SQL server 1.2 demonstrated reasons for using databases 1.3 Applied relational modeling concepts 1.4 Created an entity relationship model 1.5 Normalized database tables 1.6 Validated an ER model 1.7 Created, edited and dropped tables using SQL 1.8 Retrieved, added, removed and updated records using SQL statements 1.9 Created and queried a database from a validated ER model. 1.10 Retrieved data from several tables using joins 1.11 Applied object oriented database concepts 1.12 Selected a database type based on user requirements. 1.13 Demonstrated Object Oriented Concepts 1.14 Demonstrated designing of views and triggers in object oriented databases. 1.15 Implemented Indexing and hashing
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	1.16 Analyzed and documented the application's databases.
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical demonstration 3.3 Observation 3.4 Written test
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

DESIGN WEB APPLICATIONS

ISCED UNIT CODE: 0613 554 06A

UNIT CODE: ICT/OS/CS/CR/10/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to demonstrate web design skills. It involves understanding HTML basics, using HTML elements, demonstrating web page formatting, applying styles, understanding JavaScript basics, using JavaScript data types, using JavaScript functions and using JavaScript libraries.

ELEMENTS AND PERFORMANCE CRITERIA

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Apply HTML basics	1.1 <i>Terminologies used in HTML</i> are applied 1.2 A HTML file is created 1.3 <i>HTML core elements</i> are applied 1.4 HTML core elements are integrated
2. Use HTML elements	2.1 <i>Basic HTML elements</i> are identified 2.2 Basic HTML elements are added to a HTML document 2.3 <i>HTML Attributes</i> are applied 2.4 Attributes are added to elements
3. Demonstrate web page formatting	3.1 <i>Layout elements</i> are selected 3.2 Layout elements are added to the HTML document 3.3 <i>Layout element attributes</i> are added to the HTML document
4. Apply styles	4.1 <i>Style concepts</i> are selected 4.2 Internal styles are applied 4.3 External CSS file is created
5. Identify JavaScript basics	5.1 JavaScript principles are applied 5.2 JavaScript syntax applied 5.3 HTML element attributes are accessed using JavaScript Document Object Model (DOM) 5.4 HTML element attributes are changed using DOM
6. Apply JavaScript data types	6.1 <i>JavaScript data types</i> are applied 6.2 Operations on the data types are performed 6.3 <i>Operations on arrays</i> is performed

ELEMENT These describe the key outcomes which make up workplace function .	PERFORMANCE CRITERIA These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
7. Use JavaScript functions	7.1 Structure of a JavaScript function is applied. 7.2 A JavaScript function is created 7.3 A JavaScript function is invoked 7.4 Values are returned using functions
8. Use JavaScript libraries	8.1 Concepts of libraries are applied. 8.2 JQuery framework is utilized 8.3 Installation of JQuery is performed 8.4 Referencing of JQuery is performed 8.5 JQuery syntax is applied 8.6 <i>JQuery events</i> are implemented. 8.7 DOM Manipulation with JQuery is performed

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
1. Terminologies used in HTML may include but not limited to:	• Document • Stylesheet • Element • Attribute
2. HTML core elements may include but not limited to:	• <head> • <title> • <body> • <html>
3. Basic HTML elements may include but not limited to:	• <p> •
 • <h1>
4. Attributes may include but not limited to:	• src • alt • href
5. Layout elements may include but not limited to:	• <header> • <nav> • <section> • <footer>
6. Layout element attributes may include but not limited to:	• Class • Id • name

Variable	Range
7. Style concepts may include but not limited to:	● Background ● Padding ● Alignment ● Border
8. JavaScript data types may include but not limited to:	● Strings ● Numbers ● Booleans
9. Operations on arrays may include but not limited to:	● count () ● pop() ● push ()
10. JQuery events may include but not limited to:	● Mouse events ● Keyboard events ● Form events ● Document / window events

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

- HTML basics
- HTML elements
- Web page formatting
- Styling
- JavaScript basics
- JavaScript data types
- JavaScript functions
- JavaScript libraries

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Created a HTML document 1.2 Added attributes to HTML documents 1.3 Formatted a web page 1.4 Added styles to a web page 1.5 Demonstrated the importance of JavaScript 1.6 Use JavaScript to change HTML elements 1.7 Demonstrated event handling in JQuery
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical demonstration 3.3 Observation 3.4 Written test
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

CREATE ALGORITHMS AND DATA STRUCTURES

ISCED UNIT CODE: 0613 554 09A

UNIT CODE: ICT/OS/CS/CR/09/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to understand algorithms and data structure. It involves Understand fundamental principles of algorithms understanding fundamental concepts of data structures, linked lists, stacks and queues, search techniques and sorting techniques

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Applied fundamental principles of algorithms	1.1 Algorithms principles are identified 1.2 Algorithms characters are applied 1.3 Algorithm are written 1.4 Algorithm are Analysed 1.5 Complexities of algorithms are computed 1.6 Greedy algorithms are applied 1.7 Divide and conquer is applied
2. Apply fundamental concepts of data structures	2.1 Key concepts in data structures are applied 2.2 Arrays are created 2.3 Array insertion operations are performed 2.4 Array delete, search and update are performed 2.5 Array operations are performed
3. Utilize Linked lists	3.1 Linked lists are created 3.2 Doubly linked lists are created. 3.3 Circular linked lists are created 3.4 Basic operations for the various linked lists are applied
4. Create Stacks and Queues	4.1 Stacks and queues are applied 4.2 Stack and queue created 4.3 Basic operations in stacks are performed 4.4 Basic operations in Queue are performed 4.5 Basic operations in stacks and queue are applied
5. Utilize Search Techniques	5.1 Search is performed 5.2 Linear Search is performed 5.3 Binary Search is performed 5.4 Search techniques are applied

ELEMENT	PERFORMANCE CRITERIA
These describe the key outcomes which make up workplace function .	These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
6. Apply Sorting Techniques	6.1 Sorting is defined 6.2 <i>Categories of sorting techniques</i> are applied 6.3 <i>Types of Sorting algorithms</i> are applied 6.4 Sorting algorithms are demonstrated using C++

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
8. Complexities may include but is not limited to:	• Space • Time
9. Greedy algorithms may include but is not limited to:	• Counting coins
10. Key concepts in data structures may include but is not limited to:	• Data • Object • Type
11. Basic operations for various linked lists may include but is not limited to:	• Insertion • Deletion • Reverse • Display
12. Basic operations in stacks may include but is not limited to:	• Push • Pop
13. Basic operations in queues may include but is not limited to:	• Enqueue • Dequeue
14. Categories of sorting techniques may include but is not limited to:	• In place • Not in place • Stable • Not stable • Adaptive • Non-adaptive
15. Types of Sorting algorithms may include but is not limited to:	• Bubble sort • Insertion sort • Selection sort

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;
- Research;

Required knowledge

The individual needs to demonstrate knowledge of:

- Fundamental principles of algorithms
- Fundamental concepts of data structures
- Linked lists
- Stacks and queues
- Search techniques
- Sorting techniques

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Wrote an algorithm 1.2 Demonstrated array operations 1.3 Demonstrated basic operations for the various linked lists 1.4 Demonstrated basic operations in stacks and queues 1.5 Demonstrated search techniques 1.6 Demonstrated sorting algorithms
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning 3.2 Practical tests

	3.3 Observation 3.4 Written tests
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.

DEVELOP INFORMATION SYSTEM

ISCED UNIT CODE: 0613 554 11A

UNIT CODE: ICT/OS/CS/CR/06/6/MA

UNIT DESCRIPTION

This unit covers the competencies required to develop an information system. It involves understanding fundamentals of information systems, applying security measures to data, hardware, software in automated environment, understanding the software development process, demonstrating human computer interaction principles, understanding the VB.net programming environment and developing and testing a VB.NET application.

ELEMENT These describe the key outcomes which make up workplace function .	PERFORMANCE CRITERIA These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
1. Apply fundamentals of Information Systems	1.1 Information system fundamentals are identified 1.2 Information systems type is selected 1.3 Emerging trends in information systems are identified and applied 1.4 Information systems are recommended for different scenarios 1.5 Information system security is implemented 1.6 Mobile security is implemented 1.7 Web security is implemented
2. Apply security measures to data, hardware, software in automated environment	2.1 <i>Data security and privacy are</i> applied in accordance with the prevailing technology 2.2 <i>Security threats</i> are identified <i>and control measures</i> are applied in accordance with laws governing protection of ICT 2.3 Computer threats and crimes are detected in accordance to Information Management security guidelines 2.4 Protection against computer crimes are undertaken in accordance with laws governing protection of ICT
3. Apply the Software Development Process	3.1 Software Development Life Cycle is identified 3.2 <i>Software Development Methodologies</i> are applied 3.3 <i>Modeling techniques</i> are applied using CASE tools

ELEMENT These describe the key outcomes which make up workplace function .	PERFORMANCE CRITERIA These are assessable statements which specify the required level of performance for each of the elements. <i>(Bold and italicized terms are elaborated in the range.)</i>
4. Demonstrate Human Computer Interaction Principles	4.1 Human Computer Interaction is applied 4.2 Interface design principles are applied 4.3 Interface is designed using a design software
5. Configure the VB.NET programming environment	5.1 The .Net framework is installed 5.2 Visual Studio is installed 5.3 Features of VB.Net are configured 5.4 The IDE environment is chosen 5.5 VB.Net program structure is developed 5.6 VB.Net project is created and compiled
6. Develop and test a VB.NET application	6.1 Basic VB.Net Controls are applied 6.2 Elements of a control are applied 6.3 Basic VB.Net Controls' Properties, Methods and Events are applied 6.4 Events are handled 6.5 Forms are designed as per HCI principles 6.6 VB.Net applications are connected to a database. 6.7 VB.NET applications are tested and deployed

RANGE

This section provides work conditions to which the performance criteria apply. It allows for different work environments and situations that will affect performance.

Variable	Range
13. Types of information systems may include but not limited to:	● Transaction Processing Systems ● Management Information systems ● Decision Support systems ● Executive Information Systems ● Office Automation Systems ● Knowledge based systems ● Expert Systems
14. Data security and privacy may include but not limited to:	● Confidentiality of data ● Cloud computing ● Integrity -but-curious data surfing
15. Security and control measures may include but not limited to:	● Counter measures against cyber terrorism ● Risk reduction ● Cyber threat issues

Variable	Range
	● Risk management ● Pass-wording
16. Security threats may include but not limited to:	● Cyber terrorism ● Hacking
17. Software development methodologies may include but not limited to:	● Waterfall ● Spiral ● Rapid Application Development ● Agile
18. Modeling techniques may include but not limited to:	● Data Flow Diagrams ● ER diagrams ● Use Case Diagrams
19. Interface Design Principles may include but not limited to:	● Usability ● Learnability ● Flexibility
20. Basic VB.Net Controls may include but not limited to:	● Form ● Text Box ● Label ● Button ● List Box ● Combo Box ● Radio Button ● Check Box ● Picture Box ● Progress Bar ● Scroll Bar ● Date Time Picker ● Tree View ● List View
21. Elements of a control may include but not limited to:	● Properties ● Methods ● Events

REQUIRED SKILLS AND KNOWLEDGE

This section describes the skills and knowledge required for this unit of competency.

Required skills

The individual needs to demonstrate the following skills:

- Communications (verbal and written);
- Time management;
- Problem solving;
- Planning;
- Decision Making;

- Research;

Required knowledge

- The individual needs to demonstrate knowledge of:
 - Fundamentals of Information Systems
 - Data security and privacy
 - Computer security threats and control measures
 - Technology underlying cyber-attacks and networks
 - Cyber terrorism
 - Computer crimes
 - Detection and protection of computer crimes
 - Laws governing protection of ICT
 - Software Development Process
 - Human Computer Interaction Principles
 - VB.NET programming environment
- Developing and testing a VB.NET application

EVIDENCE GUIDE

This provides advice on assessment and must be read in conjunction with the performance criteria, required skills and understanding and range.

1.Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified types of information systems 1.2 Identified and controlled Security threats are measures are applied in accordance with laws governing protection of ICT 1.3 Applied Software Development Life Cycle 1.4 Applied Software Development Methodologies 1.5 Demonstrated Modelling techniques using CASE tools 1.6 Created a VB.NET project demonstrating event handling, form design and connection to the database
2.Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace where assessment can take place 2.2 Appropriately simulated environment where assessment can take place 2.3 Resources relevant to proposed activity or task
3. Methods of Assessment	Competency may be assessed through: 3.1 Oral questioning

	3.2 Practical tests 3.3 Observation 3.4 Written tests
4. Context of Assessment	Competency may be assessed 4.1 Off the job 4.2 on the job 4.3 During industrial attachment
5. Guidance information for assessment	5.1 Holistic assessment with other units relevant to the industry sector, workplace and job role is recommended.